

Albara Mehene

617-997-9292 | amehene95@gmail.com

github.com/FloatingAI | [linkedin.com/in/albaramehene](https://www.linkedin.com/in/albaramehene) | <https://floatingai.github.io>

Professional Experience

Software Engineer II, Canon U.S.A., Cambridge, MA

September 2023 - June 2025

- Built a C++ API enabling real-time communication between GUI interfaces and robotic hardware, enhancing system responsiveness during clinical operations.
- Architected and Integrated a service health manager written in C++ that monitors active robot diagnostics of health (temperature, voltage and currents) helping prevent system damage.
- Designed and implemented wxWidgets and C++ based GUI for controlling robot capabilities during a real-time clinical procedure.
- Spearheaded the automation of CI/CD pipelines using Jenkins, streamlining testing and deployment workflow, increasing our delivery cycles and eliminating manual build bottleneck.
- Optimized and modularized legacy codebase, enabling independent module compilation and reducing build times by 60%.

Software Engineer II, BAE Systems, Burlington, MA

January 2022 - September 2023

- Created a web application that alerts and monitors live network traffic between radio platforms to measure better metrics using React.js and PostgreSQL
- Developed new features and tools in C++ for aircraft control & a messaging system in Python used in real-time simulation runs.
- Maintained and extended software running on radio platforms using Python to translate coordinates and communicate between devices
- Refactored and decoupled core legacy components to utilize newer developer tools, resulting in over 30% reduction in build time and faster testing
- Led & migrated CI/CD environment from BitBucket to Gitlabs to gain access to newer tools and functionality such as Docker

Software Engineer I, Raytheon Technologies, Woburn, MA

June 2019 – December 2021

- Developed and maintained tools in C++ that helped improve functionality and efficiency by implementing a UI with additional features and capabilities for radar runs
- Used Python to implement scripts that insert and parse data into MySQL databases helping reduce the number of steps required for a Six Sigma project
- Implemented automation scripts to streamline the CI/CD process for Jenkins and GitHub Actions
- Presented and monitored database metrics for stakeholders through analytic tools such as Grafana
- Refactored and refined legacy code for better maintainability and in support of a migrating effort to newer technologies for the team

Education

University of Massachusetts Lowell, Lowell, MA

- B.S in Computer Science, Graduated Spring 2019 | Lowell, MA

Skills

- **Programming Languages & Libraries:** C++, Python, JavaScript, React.js, C, SQL, NoSQL, InfluxDB, HTML, CSS, GPIO APIs, IA32.
- **Technologies:** Git, Jenkins, wxWidgets, Node.js, Django, Grafana, AWS, Flask, Docker, Bitbucket, IoT
- **Operating Systems:** Windows, Linux (Ubuntu, Debian, RedHat)